UAP Nexus AI: Intelligent Academic Assistant for University of Asia Pacific

Maisha Noor
22201230
Dept. of CSE
University of Asia Pacific
22201230@uap-bd.edu

Udoy Titus Rebeiro
22201224
Dept. of CSE
University of Asia Pacific
22201224@uap-bd.edu

Ekra Chowdhury Risha
22201246
Dept. of CSE
University of Asia Pacific
22201246@uap-bd.edu

***Abstract*—Managing university information remains a continual challenge, with students often spending excessive time sifting through extensive PDF files, while staff are burdened by a high volume of repetitive questions. This document introduces UAP Nexus AI, an advanced academic chatbot based on Retrieval-Augmented Generation (RAG), created for the University of Asia Pacific (UAP). The system processes university PDF files, prepares and segments the text, creates dense vector embeddings with the intfloat/e5-large-v2 sentence transformer model, and organizes them using FAISS for efficient similarity-based retrieval. Answer generation utilizes the `google/flan-t5-base` model, which is deployed as a functioning REST API through Flask and ngrok. The evaluation of the system was conducted on 60 ground-truth question-answer pairs utilizing Semantic Similarity, Retrieval Accuracy, BLEU, and ROUGE metrics, resulting in scores of 0.8379, 0.78, 0.0693, and 0.4003 respectively. Findings show that the system consistently obtains institution-specific data and produces responses that are friendly for students, based solely on validated university documents, greatly minimizing the risk of hallucination.**

***Index Terms*—Retrieval-Augmented Generation, Dense Retrieval, FAISS, Flan-T5, Natural Language Processing, University Chatbot, Sentence Transformers**

I. Introduction

The increase in digital university documents has led to an issue with information accessibility for students. Though institutions keep detailed records on admissions, tuition, academic policies, exam schedules, and contact details, manually finding specific answers in these documents is laborious and ineffective. The management of repetitive student inquiries by staff is also sluggish and inconsistent, leading to a diminished student experience and increased administrative burden. Recent developments in Natural Language Processing (NLP), especially in dense retrieval and large language models, have created new avenues for constructing domain-specific question-answering (QA) systems that function on private institutional databases without depending on public internet data [1]. Retrieval-Augmented Generation (RAG), presented by Lewis et al. [2], merges the advantages of information retrieval with sequence-to-sequence text generation, enabling a system to base its responses on retrieved evidence instead of depending exclusively on parametric model knowledge. This method is particularly appropriate for institutional settings where precision, reliability, and subject matter expertise are essential. This document introduces UAP Nexus AI, a chatbot based on RAG, created specifically for the University of Asia Pacific. The system addresses student questions by extracting pertinent sections from a selected knowledge base of university PDF files and producing natural language answers with the Flan-T5 model [3]. The main goals of the project are listed below:

- To create an automated academic information system for UAP students that is accessible 24/7.
- To establish a hybrid retrieval system that merges FAISS vector searching with re-ranking based on cosine similarity.
- To assess the system thoroughly employing conventional NLP metrics such as BLEU [4], ROUGE [5], semantic similarity, and retrieval precision.
- To implement the system as a REST API that is publicly available for practical utilization.

The system fulfills a real societal requirement: students can receive immediate, precise answers to university-related questions at any hour, while administrative personnel are freed from the task of managing repetitive information inquiries.

II. Dataset and Preprocessing

A. Knowledge Base (Source 1)

The main knowledge base is made up of five official PDF documents from the University of Asia Pacific. These documents include information on admissions, policies for tuition fees and waivers, the university's vision and mission statements, exam schedules, and contact details. Collectively, they encompass the most frequently requested types of university information by students. The `pypdf` library was utilized to extract text from PDFs. Preprocessing entailed changing all text to lowercase, eliminating URLs, special characters, and unnecessary whitespace to minimize noise in the retrieved context. Abnormal documents characterized by unusually short or lengthy word counts were detected and removed via the Interquartile Range (IQR) method, a conventional statistical approach for identifying outliers [6]. Boxplot examination of document lengths verified that the resulting corpus preserved a uniform length distribution appropriate for chunking.

B. Question-Answer Dataset (Source 2)

For evaluation purposes, a dataset of 60 ground-truth question-answer pairs was manually compiled in JSON format from the university documents. Every entry includes a question, its related reference answer obtained from university documents, and relevant metadata. This dataset was solely utilized as the evaluation benchmark and was not employed in building the retrieval index.

C. Document Chunking

Documents were divided into segments utilizing LangChain's `RecursiveCharacterTextSplitter` [7], featuring a chunk size of 350 words and an overlap of 80 words. The overlap guarantees that contextual continuity is maintained across chunk boundaries, which is vital for obtaining comprehensive answers to inquiries whose pertinent content extends over several paragraphs. This chunking approach aligns with best practices in the RAG literature [2], which advocate for chunk sizes that suitably balance retrieval detail with adequate contextual details for the generator.

III. System Design and Methodology

The UAP Nexus AI framework is structured into five phases: Data Loading, Document Processing, Retrieval System, Answer Generation, and Web API Deployment. The complete pipeline adheres to a conventional RAG architecture [2], incorporating domain-specific design choices at every step illustrated in Fig. 1.

A. Embedding Model

Text segments are transformed into compact vector representations through the `intfloat/e5-large-v2` sentence transformer model [8]. This model generates 384-dimensional embeddings and provides excellent semantic similarity capabilities for retrieval tasks, all while being computationally efficient for deployment in the GPU environment of Google Colab. During the evaluation phase, the `all-MiniLM-L6-V2` model [9] was also used to produce high-quality embeddings for evaluating semantic similarity between generated responses and reference answers.

B. FAISS Retrieval Index

All chunk embeddings are kept in a FAISS `IndexFlatL2` index [10], which executes precise nearest-neighbor searches utilizing L2 (Euclidean) distance. Once a user sends a query, it is encoded with the same sentence transformer model, and the four most similar chunks are fetched from the index. FAISS was chosen for its computational efficiency and scalability, enabling billion-scale similarity searches with GPU acceleration [10].A hybrid scoring mechanism for retrieval was developed to enhance result quality beyond just FAISS retrieval.The hybrid score is calculated as:

$$S_{hybrid} = 0.7 \times S_{cosine} + 0.3 \times S_{FAISS} \quad (1)$$

where S_{cosine} represents the cosine similarity metric and S_{FAISS} denotes the normalized inner product score from FAISS. This formula places greater emphasis on semantic cosine similarity while also considering FAISS scores as a secondary indicator. The top-4 chunks that have been re-ranked are utilized as context for generating answers. The retrieved context is cut off at 250 words to fit the input length limitations of the Flan-T5-base model.

C. Answer Generation with Flan-T5

The component for generating answers utilizes the `google/flan-t5-base` model [3], [11], a sequence-to-sequence transformer with 990 million parameters that is instruction-tuned. The model is given a structured prompt that includes the retrieved context along with the user's query, organized as: "Answer the following question based only on the provided context. Context: [retrieved chunks]. Question: [user query]. Answer:"

This method of prompt engineering clearly confines the model to the given context, minimizing hallucination and ensuring that all answers are based on validated university documents [12]. A backup answer is provided when retrieval certainty is minimal.

D. Web API Deployment

The chatbot is implemented as a Flask REST API with a `POST /chat` endpoint that takes a JSON body with the user's question and provides the generated response. The API is made publicly accessible through `pyngrok` tunneling, allowing integration with the frontend interface of the UAP university website, which is constructed using HTML, CSS, and JavaScript.

IV. Implementation

The suggested UAP Nexus AI system was developed using Google Colab. The execution adopts a modular pipeline structure, where each phase is individually verifiable and jointly creates a cohesive end-to-end RAG system covering 46 notebook cells.

A. Environment Setup and Libraries

All experiments were performed on Google Colab utilizing a T4 GPU runtime for enhancement. Both inference based on transformers and generation of embeddings. Table Ioutlines the essential libraries utilized throughout the implementation.

B. Building a Knowledge Base

Text was retrieved from every PDF file utilizing `pypdf` and processed through a preprocessing pipeline where every character was changed to lowercase, URLs and special characters were eliminated using regular expressions, and files with Anomalous lengths were removed utilizing the Interquartile Range (IQR) technique. [6]. The processed text was divided into overlapping segments. Utilizing LangChain's `RecursiveCharacterTextSplitter`, generating an overall total of 349 segments throughout the five documents. Every segment was transformed into a 384-dimensional embedding vector using `all-MiniLM-L6-v2` and saved in a FAISS`IndexFlatL2` structure for later retrieval.

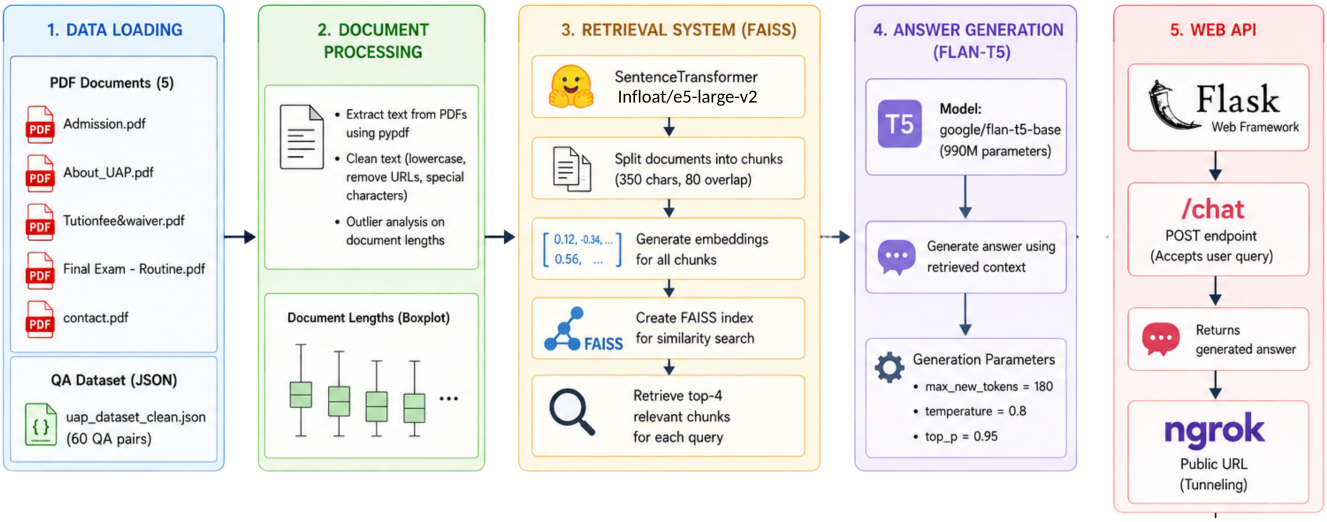


Fig. 1. End-to-end architecture of the UAP Nexus AI system, comprising five sequential stages.

TABLE I
LIBRARIES AND FRAMEWORKS USED IN SYSTEM IMPLEMENTATION

Library / Framework	Purpose
pypdf	PDF text extraction and document parsing
langchain_text_splitters	Document chunking and context segmentation
sentence-transformers	Dense semantic embedding generation
faiss-cpu	Vector indexing and similarity retrieval
transformers (HuggingFace)	Transformer model loading and inference
google/flan-t5-base	Natural language answer generation
Flask	REST API backend and chatbot service deployment
pyngrok	Secure public URL tunneling for deployment
matplotlib, seaborn, pandas	Data analysis and performance visualization

C. API Deployment and Frontend Integration

The entire pipeline is contained within a Flask REST API that offers a single`POST /conversation` endpoint. The endpoint receives a JSON request body that includes the user's query in natural language and provides a JSON response containing the generated response. The Flask server is made available to the public through `pyngrok`, which associates the local Colab runtime port with a permanent public HTTPS URL. This link is incorporated into the UAP university site's frontend, created using HTML, CSS, and JavaScript, allowing students to engage with the chatbot directly from the college internet platform. A function for post-processing (`clean_output()`) isused on the raw model output to standardize casing, eliminate artifacts, and guarante equality of the response prior to providing the answer to the client.

V. RESULTS AND ANALYSIS

The UAP Nexus AI system was assessed using 60 factual question-answer pairs, extracted from the hand-selected JSON dataset. Performance was evaluated using four complementary measures—Semantic Similarity, Retrieval Precision, ROUGE, and BLEU scores—to reflect both semantic accuracy and lexical conformity. Intersection between created and standard responses. The comprehensive assessment findings are shown in Table II. As illustrated in Table II, the system attained a semantic similarity score of 0.8379, calculated as the cosine similarity involving `e5-large-v2` embeddings [8] of the produced and source responses. This rating suggests that the system reliably generates answers that are semantically near the actual truth, even when the surface phrasing varies, which is the anticipated operation of a generative QA system that rephrases retrieved content instead of replicating it word for word. The accuracy of retrieval stands at 0.78, adding to the evidence that the hybrid FAISS retrieval pipeline effectively brings to light pertinent context segments in most question scenarios, along with the accurate response found within the leading four retrieved segments for 78% of the test inquiries. The ROUGE score of 0.4003 [5] indicates a moderate level of lexical overlap. Among the produced and benchmark responses. This outcome aligns with anticipations for open-domain generative QA systems, in which models restate data from obtained context instead of taking it word-for-word, fundamentally restricting n-gram overlap using predefined reference strings. Likewise, the BLEU metric of 0.0693 [4] is typically low, a trend commonly documented in the RAG literature [2], illustrating the adaptability of natural language generation, which creates syntactically varied results that differ

from the precise n-gram arrangement of reference responses. When combined, the difference the connection between the elevated semantic similarity and the reduced lexical scores indicate that the system generates semantically accurate replies articulated in diverse natural language, which is the intended function for a student-oriented academic helper.

TABLE II
EVALUATION RESULTS OF UAP NEXUS AI ON 60 GROUND-TRUTH QA PAIRS.

Metric	Score
Semantic Similarity	0.8379
Retrieval Accuracy	0.7800
ROUGE Score	0.4003
BLEU Score	0.0693

A. Answer Quality Classification

To better define system performance at a more detailed level, produced responses were categorized into one of four quality levels according to their distinct semantic similarity ratings: Outstanding (≥0.90), Fair (0.75–0.89), Incomplete (0.60–0.74), and Low ($¡ 0.60). Fig. 3 displays the bar graph of each of the four RAG evaluation metrics, presenting a visual overview of the comparative performance throughout the assessment criteria detailed in Table II. The spread of semantic similarity scores for each question is additionally shown in the histogram displayed in Fig. 2. As shown in Fig. 2, the similarity score distribution is primarily right-skewed, with the majority of responses focused in the 0.65–0.70 interval and another notable group in the 0.90–1.00 interval. This bimodal pattern indicates that the system either acquires very pertinent context and produces a superb response or fetches somewhat related context, leading to a diminished reaction, with notably fewer instances in the intermediate span.

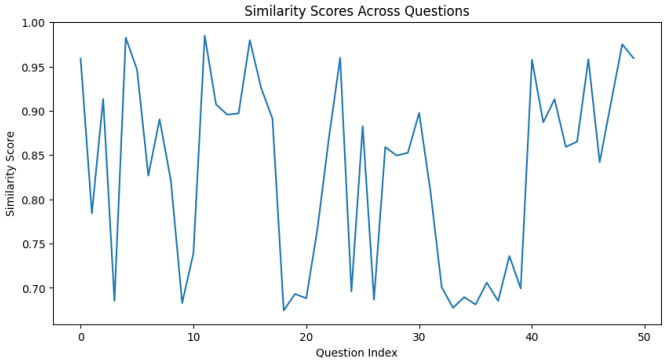


Fig. 2. Histogram showing the distribution of semantic similarity scores across the 60 test question-answer pairs.

The bar chart in Fig. 3 reinforces the idea that Semantic Similarity and retrieval accuracy are the most effective metrics, visually. supporting the numerical results presented in Table II.

Qualitative examination of specific test cases supports these conclusions. the inquiry *"What is the process to apply to UAP?"*, the system provided a reply with a similarity score of 0.9591 (Excellent), precisely reflecting the online process of applying outlined in the admissions document. *"Which documents are needed for entry to UAP?"* The system accurately recognized SSC and HSC transcripts and certificates from the necessary papers, attaining a score of 0.9134 (Outstanding). Conversely, for the inquiry *"Is there a test for admission at UAP?"*, the model replied simply with ‘ ”no” in comparison to a more detailed reference answer, resulting in a lower similarity score of 0.6771 (positive). This instance of failure illustrates the vulnerability of smaller generative models to binary factual inquiries where the reference response includes context expansion beyond a straightforward yes or no answer [3], [11].

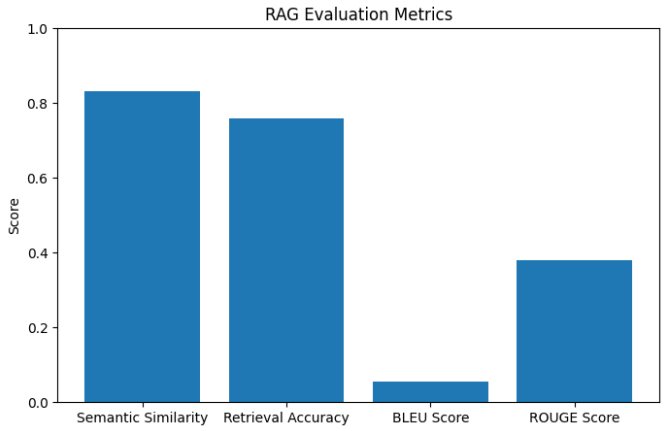


Fig. 3. Bar chart of RAG evaluation metrics: Semantic Similarity, Retrieval Accuracy, ROUGE Score, and BLEU Score.

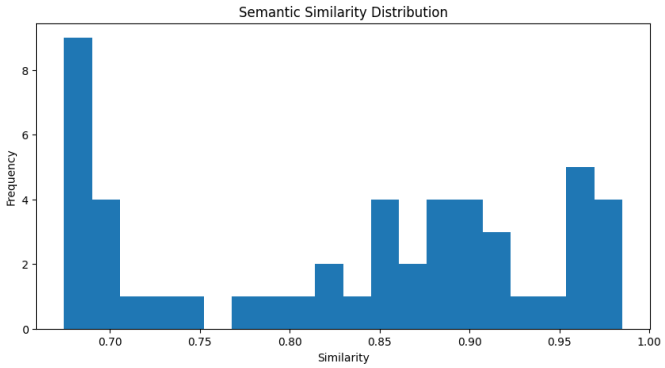


Fig. 4. Line plot of semantic similarity scores across question indices, illustrating per-question performance variation over the test set.

The line graph in Fig. 4 demonstrates the fluctuation of semantic similarity ratings throughout all question indexes in the evaluation set. Substantial variability is evident among the 50 assessed questions, with scores varying from around 0.65 to 1.00.

This variance is not evenly distributed but instead seems fragmented, featuring groups of consecutive queries that perform well scattered with solitary droplets. Such a trend indi-

cates that effectiveness degradation is confined to particular types of queries—notably brief, unclear,

B. Feature Correlation Analysis

To examine the connection between structural characteristics of the QA pairs and system efficiency, a Pearson correlation analysis was performed on three variables: the length of the question (in words), the length of the answer (in words), and the semantic similarity index. The correlation matrix produced and its heatmap representation are shown in Fig. 5.

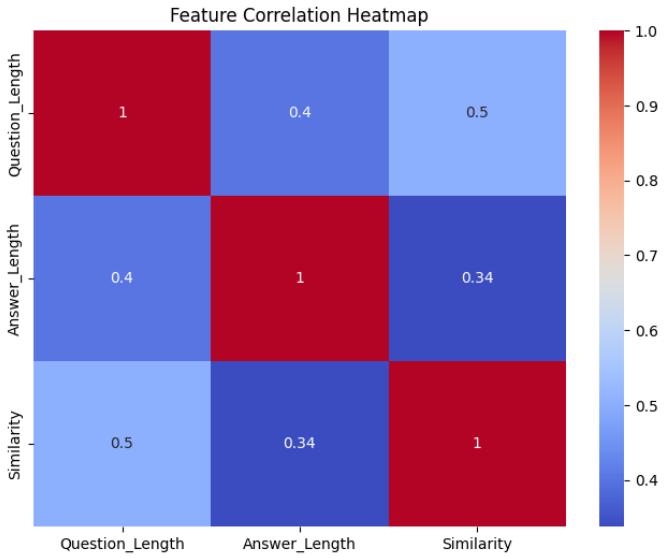


Fig. 5. Pearson correlation heatmap of question length, answer length, and semantic similarity score, with correlation coefficients annotated in each cell.

As shown in Fig. 5, the length of questions and their semantic similarity show a slight positive correlation (r = 0.38), whereas response length and semantic similarity demonstrates a comparably weak positive correlation (r = 0.37). Length of the question and the length of the responses show a moderate correlation (r = 0.40), indicating longer inquiries often result in more extensive generated replies. The lack of robust associations between all three variables, evidently apparent in the off-diagonal cells of Fig. 5, shows that no individual structural characteristic is a primary determinant of response quality, with each variable adding independently from the noticed differences in similarity scores. The evident positive relationship between the length of questions and their similarity found in Fig. 5 aligns with results observed in the dense retrieval literature [13], in which lengthier and more detailed queries offer a more detailed semantic signal for the embedding model, enhancing accuracy of gathered segments. On the other hand, brief or vaguely defined queries generate embedding vectors that are not as discriminative inside the FAISS index, enhancing the probability of obtaining slightly relevant context and thereby diminishing the creation standard. This observation directly impacts the system. Motivating users to create more precise inquiries or executing a query expansion component [1] to automatically enhance brief queries may result in significant improvements in retrieval accuracy and overall response caliber.

VI. Discussion

This document came out with UAP Nexus AI that effectively resolves the issue of ineffective university information retrieval through the integration of dense vector embeddings, FAISS-driven hybrid retrieval, and Flan-T5 generative answering into a fully deployed chatbot accessible via API. Evaluation outcomes indicate robust semantic performance—with a semantic similarity of 0.8379 and a retrieval accuracy of 0.78—confirming that the system consistently bases its responses on verified institutional documents.

VII. Conclusion

This document introduced UAP Nexus AI, an intelligent academic assistant based on RAG for the University of Asia Pacific. The solution effectively tackles the challenge of ineffective university information retrieval by integrating dense vector embeddings, FAISS-driven hybrid retrieval, and Flan-T5 generative responses into a completely deployed, API-enabled chatbot. Evaluation findings show robust semantic performance — a semantic similarity of 0.8379 and a retrieval accuracy of 0.78 — and affirm that the system consistently bases its responses on validated institutional documents. The main drawback is the generative capability of the Flan-T5-base model when handling complex or ambiguous questions. Future efforts will aim at substituting Flan-T5 with a more extensive language model like LLaMA or GPT to enhance answer fluency and the depth of reasoning. Future enhancements aim to broaden the PDF knowledge base to encompass more university departments and services, introduce multi-turn conversation capabilities with session memory, and create a specific frontend user interface for students to improve accessibility.

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